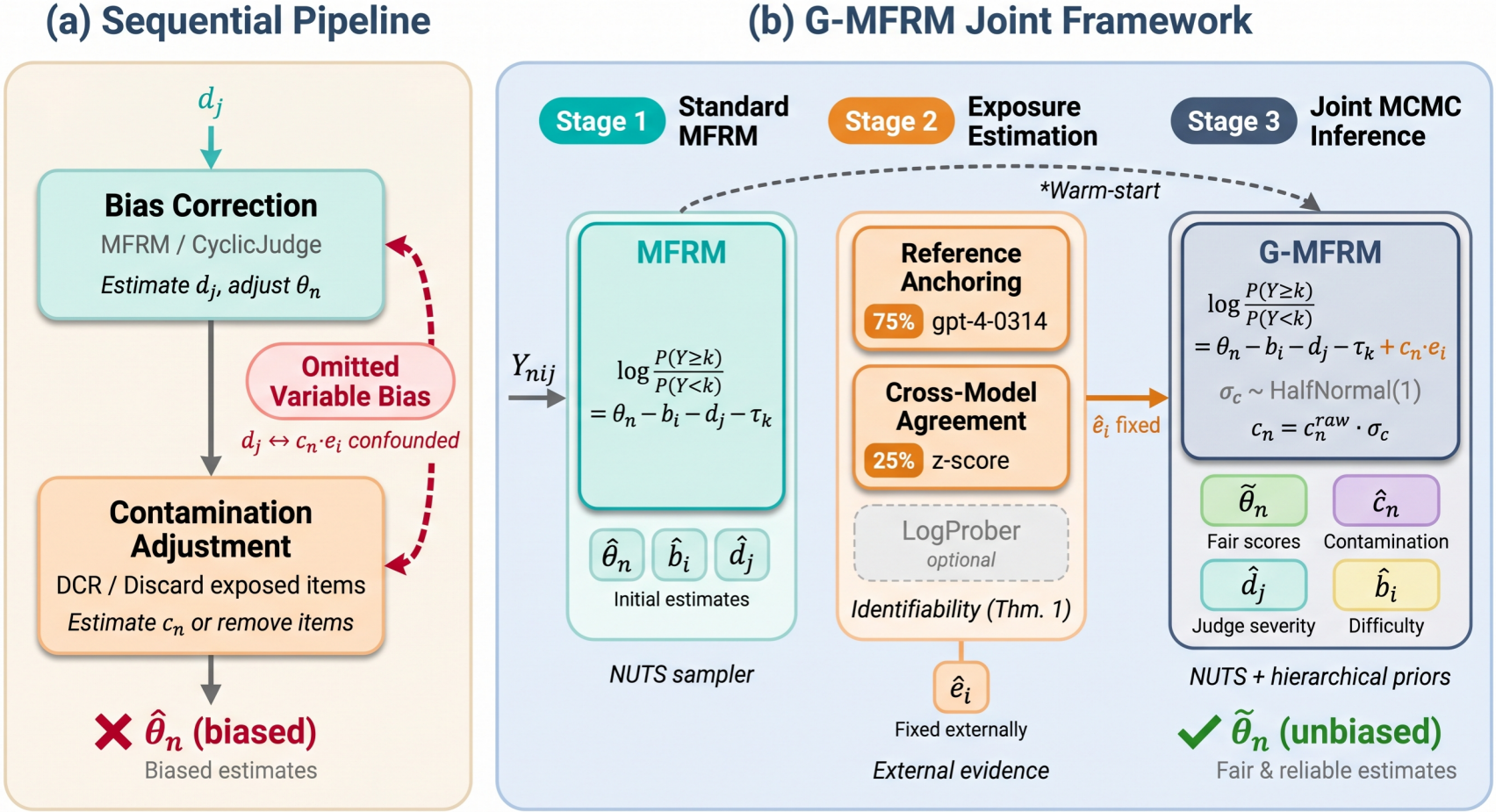

(a) Sequential Pipeline
d_j
Bias Correction
MFRM / CyclicJudge
Estimate d_j, adjust θ_n
Omitted Variable Bias
$d_j \leftrightarrow c_n \cdot e_i$ confounded
Contamination Adjustment
DCR / Discard exposed items
Estimate c_n or remove items
$\hat{\theta}_n$ (biased)
Biased estimates
(b) G-MFRM Joint Framework
Stage 1 Standard MFRM
Stage 2 Exposure Estimation
Stage 3 Joint MCMC Inference
*Warm-start
Y_{nij}
MFRM
$\log \frac{P(Y \geq k)}{P(Y < k)} = \theta_n - b_i - d_j - \tau_k$
$\hat{\theta}_n$ $\hat{b}_i$ $\hat{d}_j$
Initial estimates
NUTS sampler
Reference Anchoring
75% gpt-4-0314
Cross-Model Agreement
25% z-score
LogProber
optional
Identifiability (Thm. 1)
$\hat{e}_i$
Fixed externally
External evidence
$\hat{e}_i$ fixed
G-MFRM
$\log \frac{P(Y \geq k)}{P(Y < k)} = \theta_n - b_i - d_j - \tau_k + c_n \cdot e_i$
$\sigma_c \sim$ HalfNormal(1)
$c_n = c_n^{raw} \cdot \sigma_c$
$\tilde{\theta}_n$ Fair scores
$\hat{c}_n$ Contamination
$\hat{d}_j$ Judge severity
$\hat{b}_i$ Difficulty
NUTS + hierarchical priors
$\tilde{\theta}_n$ (unbiased)
Fair & reliable estimates